Complex Numbers: Algebra

Questions 1–48

Question 1. Write the following as imaginary numbers:

a) $\sqrt{-9}$
b) $\sqrt{-2}$
c) $2\sqrt{-4}$
d) $-3\sqrt{-2}$

Question 2. Simplify $\sqrt{-a} \times \sqrt{-b}$ where a and b are real positive numbers.

Question 3. Write down the simplified form of each of the following:

a) i^2
b) i^3
c) i^4
d) i^5
e) i^{40}
f) i^{41}
g) i^{99}

Question 4. Write down the simplified form of each of the following, where n is a positive integer:

a) i^{4n}
b) i^{4n+1}
c) i^{4n+2}
d) i^{4n+3}
e) i^{4n+17}
f) i^{4n+26}

Question 5. Determine the sum of the expression

$$i + i^2 + i^3 + \cdots + i^{1000}$$

Fully justify your answer.

Question 6. Simplify

a) $(1-i)^2$
b) $(1-i)^3$
c) $(1-i)^4$
d) $(1+i)^2$
e) $(1+i)^3$
f) $(1+i)^4$

Question 7. Without using a calculator, simplify each of the following:

a) $\sqrt{-12} \times \sqrt{-27}$

b) $(2i)^3$

c) $i^2(i+3)$

d) $i + 4i^5$

e) $(i - 3i^3)(1 - 2i^4)$

f) $(2i^6 - i^9)^2$

Question 8. Evaluate

$$\left(i^{12} + \left(\frac{1}{i}\right)^{18}\right)^3$$

Question 9. Evaluate

$$\left(i^{30} - \left(\frac{1}{i}\right)^{12}\right)^4$$

Question 10. Given that $z_1 = 2 - 3i$, $z_2 = 1 + i$ and $z_3 = 5 - 2i$ find each of the following:

a) $z_1 + z_2$
b) $z_1 + z_3$
c) $z_3 - z_1$
d) $2z_2 - z_1$
e) $5z_1 + 3z_2$

Question 11. Show that $(2+3i)^2 = -5+12i$

Question 12. Show that $(-1+5i)^3 = 74-110i$

Question 13. Given x and y are real numbers, find the real and the imaginary parts of each of the following:

a) $z = 2 + (3x - iy)$

b) $z = (x - 2i) - (2x + 3iy)$

c) $z = 5(x - iy) + 2(x + iy)$

Question 14. Find the value(s) of real numbers x and y that satisfy each of the following equations:

a) $5x + 2y + (3x + 4y)i = 25 + 29i$

b) $2x + 8y + (x - 4y)i = -20 + 14i$

c) $(-3x + 2ix) + (2y + 7iy) = -14 - 74i$

d) $x + 3x^2 + (9x - y)i = 2 + i$

Question 15. Find the value of real number a such that

$$(3+7i)(a+2i)$$

is purely imaginary.

Question 16. Given that

$$(2+i)(a+bi)$$

is purely imaginary, simplify the expression giving your answer only in terms of a

Question 17. Given that

$$(a - bi)(a + 3bi) = 52 + 16i$$

where a and b are real integers, find the possible values of a and b

Question 18. Write each of the following in the form $z^2 + pz + q$ where p and q are complex numbers:

a) $(z + 1 - 2i)^2$

b) $(z - 2 + 3i)^2$

c) $(z + 1 + 2i)(z - 2 + 3i)$

Question 19.

a) Simplify $(7+2i)(3+8i)$

b) Simplify $(7-2i)(3-8i)$

c) It is given that $5^2+62^2=3869$. Using this and your answer to part a) and part b), find the prime factorisation of 3869

Question 20. Determine the possible values of z for each of the following:

a) $z^2=5+12i$

b) $z^2=21-20i$

c) $z^2=\frac{3}{16}-\frac{1}{4}i$

Question 21. Find, giving your answer in exact form:

a) $\sqrt{4+3i}$

b) $\sqrt{-12+16i}$

c) $\sqrt{2+4\sqrt{2}i}$

Question 22. Solve the simultaneous equations below to find complex numbers w and z:

$$2iz + 4w = -6 + 8i$$
$$-3z + 5wi = 7 - i$$

Question 23. For each of the following, find the possible value(s) of the unknown real value(s) a:

a) $\text{Re}[(1+ai)^3] = 0$
b) $\text{Im}[(3+ai)^3] = 0$
c) $\text{Im}[(4+ai)^4] = 0$
d) $\text{Re}[(2-ai)^4] = 0$
e) $\text{Im}[(1-ai)^5] = 0$

Complex Conjugates

Question 24. Write down the complex conjugate, z^* for each of the following:

a) $z = 2 + i$
b) $z = 3 - i$
c) $z = -6 - i$
d) $z = 1 + 5i$
e) $z = -4 - 2i$
f) $z = \frac{1}{2} + \frac{3}{4}i$
g) $z = \sqrt{5} - \sqrt{3}i$
h) $z = 2i$
i) $z = -10$

Question 25. Given the complex number $z = a + bi$, prove that each of the following are real numbers:

a) $z + z^*$

b) zz^*

Question 26. Given $w = 2 - i$ and $z = -5 + 3i$, find:

a) $2w - z$

b) $w^* + w$

c) $2w^* + z$

d) z^2

e) wzz^*

f) $w(z^*)^2$

g) $(w^*)^3$

Question 27. Given $z = x + iy$

a) Find the real and imaginary parts of $4iz + 3z^*$

b) Hence find the values of x and y such that $4iz + 3z^* = 5 - 3i$

Question 28. Find the complex number z if

a) $2z^* = 3 - 4i$

b) $3z^* + 1 = 8i$

c) $iz^* = -1 + i$

Question 29. Find z given

a) $z + 2z^* = 3 + 9i$

b) $2z - z^* = -4 - 5i$

c) $z + 5z^* = -i$

Question 30. Prove that

a) $(z^*)^2 = (z^2)^*$

b) $(z^*)^3 = (z^3)^*$

Question 31. Given the complex number $z = a + bi$, where a and b are real numbers, find

a) $z - z^*$
b) $(z - z^*)^2$
c) $(z - z^*)^3$
d) $(z - z^*)^4$
e) $(z - z^*)^{18}$

Question 32. Find the four solutions to the equation

$$(4z - z^*)^* = z^2$$

Question 33. The two complex numbers z and w satisfy

$$2z + 4z^* = 18 + 4i$$
$$5w - 3z^* = -3 + 2i$$

Find the value of w in the form $a + bi$

Question 34. A complex number z satisfies the equation,

$$z^2 + 3iz^* = 8$$

where $\mathrm{Re}(z) < 0$. Find z in the form $a + bi$ where a and b are real constants to be found.

Question 35. A complex number z satisfies the equation,

$$z^2 + 4z^* = -4 + 10i$$

where $\mathrm{Re}(z) > 0$. Find z in the form $a + bi$ where a and b are real constants to be found.

Rationalising the Denominator

Question 36. Without using a calculator, simplify each of the following:

a) $\frac{4}{i}$
b) $\frac{7}{1+i}$
c) $\frac{9}{5-2i}$
d) $\frac{8}{2+3i}$
e) $\frac{2+i}{4-i}$
f) $\frac{(1-i)^2}{1+i}$
g) $\frac{a+bi}{b-ai}$

Question 37. Given that $z = -1 + 2i$, express $z + \frac{5}{z}$ in the form $a + bi$ where a and b are real constants to be found.

Question 38. Ohm's law states that $V = IR$ where I is the current through the conductor (amps), V is the voltage measured across the conductor (volts), and R is the resistance of the conductor (ohms).

a) Given an electrical circuit has a current of $(3 - 2i)$ amps and a resistance of $(4 + i)$ ohms, find the voltage for this circuit.

b) Given the voltage on a circuit is $(6 - 2i)$ volts and a current of $(2 + 3i)$ is passed through the conductor, calculate the resistance of the conductor giving your answer in its simplified form.

Question 39. Find z given that

$$\frac{15-5i}{z} = 2-3i$$

Question 40. Evaluate

$$\frac{2}{3+i} + \frac{3}{2+i}$$

Question 41. Express

$$\frac{1}{\cos\theta - i\sin\theta}$$

in the form $a + bi$

Question 42. Given x and y are real, find the real and the imaginary parts of each of the following:

a) $z = x + iy + \frac{1}{x+iy}$
b) $z = x + iy - \frac{1}{x+iy}$
c) $z = x - iy + \frac{1}{x-iy}$
d) $z = x - iy - \frac{1}{x-iy}$

Question 43. Find the least positive integer of n for which

$$\left(\frac{1+i}{1-i}\right)^n = 1$$

Question 44. Find the values of integers a and b such that

$$\frac{a}{1-2i} + \frac{b}{3-2i} = \frac{-5+6i}{1+8i}$$

Question 45. Given $z = a + bi$, and that

$$zz^* = 17 \qquad \frac{z}{z^*} = \frac{15}{17} - \frac{8}{17}i$$

determine the possible values of a and b

Question 46. Given the complex number $z = a + bi$, find the real and imaginary parts of

$$\frac{z}{z-2}$$

Question 47. Find z given that

$$\frac{z}{3+4i} + \frac{z-1}{5i} = \frac{5}{3-4i}$$

Question 48. Given that

$$x - iy = \sqrt{\frac{a-bi}{c-di}}$$

where a, b, c and d are real numbers, show that

$$x^2 - y^2 = \frac{ac+bd}{c^2+d^2}$$

Complex Numbers: Quadratic Equations

Questions 1–12

Question 1. Bruce is attempting to find the solutions to an equation. His attempt is shown below.

$$\begin{aligned} z^2 - 2z + 5 &= 0 \\ (z-2)^2 - 4 &= -5 \\ z - 2 &= \sqrt{-9} \\ z &= 2 \pm 9i \end{aligned}$$

a) Identify all his mistakes.
b) Write a correct solution.

Question 2. Without using a calculator, find the solutions to each of the following equations. You must show your working:

a) $z^2 + 1 = 0$
b) $z^2 + 10 = 0$
c) $z^2 - 2z + 17 = 0$
d) $4z^2 - 4z + 17 = 0$
e) $2z^2 - 2z + 1 = 0$
f) $z^2 - z + 3 = 0$

Question 3. Find the solutions to each of the following, giving your answer in the form $a + bi$:

a) $(z + 5)^2 = -7$

b) $(z - 1)^2 = -12$

c) $2(z + 4)^2 + 8 = 6$

d) $8(z + 5)^2 - 3 = -9$

Question 4. Without using a calculator, find all the solutions to

$$z^4 + 2z^2 - 3 = 0$$

You must show your working.

Question 5. Find the quadratic equation in the form

$$z^2 + bz + c = 0$$

which has roots:

a) $1 + i$ and $1 - i$
b) $3 + i$ and $3 - i$
c) $1 + 2i$ and $1 - 2i$
d) $-4 + 3i$ and $-4 - 3i$
e) i and $-i$
f) $5i$ and $-5i$
g) $-1 + \sqrt{2}i$ and $-1 - \sqrt{2}i$

Question 6. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $3 - 8i$, where p and q are real values.

a) Write down the other root of the equation.
b) Find the values of p and q

Question 7. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $-\frac{1}{2} - 4i$, where p and q are real values.

a) Write down the other root of the equation.

b) Find the values of p and q

Question 8. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $a + bi$, where p, q, a and b are real values.

a) Write down the other root of the equation.

b) Find the values of p and q in terms of a and b

Question 9. Prerequisite knowledge for Q9: Complex Loci

The function f is defined as

$$\mathrm{f}(z) = z^2 - 4z + 7$$

a) Find the roots of $\mathrm{f}(z) = 0$

The two of the roots of $\mathrm{f}(z) = 0$, z_1 and z_2 lie on the locus $C = \{z : |z| = p\}$

b) Find the value of p

c) Find the area of the region in the Argand diagram enclosed by the locus C

Another point w lies on the locus C such that $\mathrm{Re}(w) = -\mathrm{Re}(z_1)$

d) State the possible values of w

The function g is defined as

$$\mathrm{g}(w) = w^2 + aw + b$$

where a and b are real constants.

e) Using your answers to part d) as the roots of $\mathrm{g}(w) = 0$, determine the values of a and b

Question 10. The equation

$$z^2 + kz + 4 = 0$$

where k is a real number, has complex roots. Find the range of possible values of k

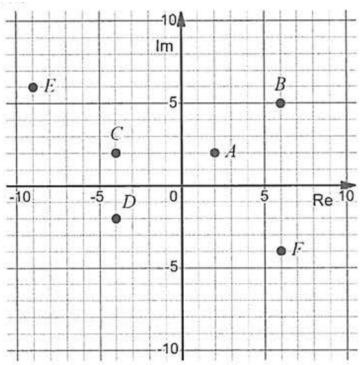
Question 11. It is given that a quadratic polynomial has non-zero real coefficients, and the linear term is zero. Show that it has purely imaginary roots.

Question 12. It is given that a quadratic polynomial has non-zero real coefficients, and the constant term is zero. Show that it can only have real roots.

Complex Numbers: Argand Diagrams

Questions 1–27

Question 1. Write down the complex numbers represented by each of the points in the Argand diagram below:



Question 2. Represent each of the following on an Argand diagram. Use a separate diagram for each part:

a) $z = 2 - 3i$ and $w = -5 + 2i$
b) $z = 3i$ and $w = 1$
c) $z = -4 + i$ and $-z$
d) $z = -1 - i$ and z^*
e) $z = 1 - 3i$, $2z$ and iz

Question 3. Represent the solution of each of the following quadratic equations on an Argand Diagram:

a) $z^2 + 2z + 4 = 0$
b) $z^2 - 3z + 3 = 0$
c) $z^2 - z + 5 = 0$
d) $z^2 + 14z + 80 = 0$

Question 4. Given that $z = 2 + i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 5. Given that $z = 3 - i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 6. Given that $z = -5 - 2i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 7. Given that $z = 1 + 3i$ and $w = 2 + 2i$, on a single diagram plot each of the following:

a) z
b) w
c) $z + w$
d) $z - w$

Question 8. Given that $z = -4 + 5i$ and $w = 5 - 2i$, on a single diagram plot each of the following:

a) z
b) w
c) $z + w$
d) $z - w$

Question 9. Given that $z = 3 + i$

a) Show that $z^2 = 8 + 6i$

b) Plot z and z^2 on the same Argand diagram.

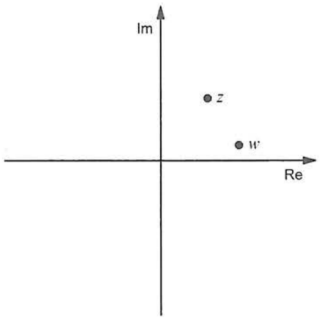
Question 10. Given that $z = -1 + i$

a) Find z^2, z^3, z^4 and z^5

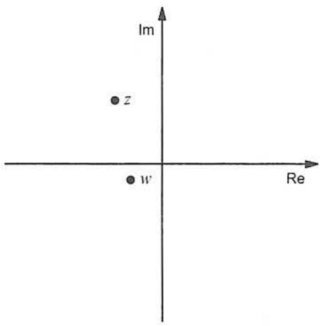
b) Plot z, z^2, z^3, z^4 and z^5 on the same Argand diagram.

Question 11. For each of the following diagrams, copy the diagram and add the points z^*, $-w$ and $z+w$

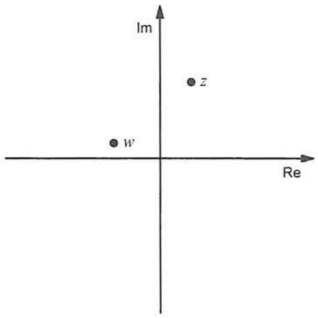
a)



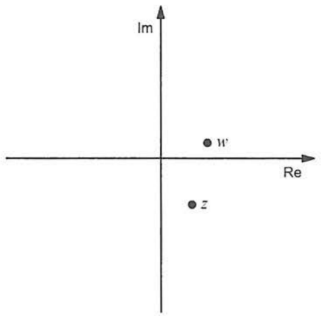
b)



c)



d)



Question 12. Prerequisite knowledge for Q12: Complex Conjugates

Given that $z_1 = 4 - 2i$ and $z_2 = \frac{1+i}{z_1}$

a) Find z_2 in the form $a + bi$

b) Show on an Argand diagram two points representing z_1 and z_2

Question 13. Given that $z_1 = 2 + i$, $z_2 = iz_1$, $z_3 = iz_2$, $z_4 = iz_3$ and $z_5 = iz_4$

a) plot z_1, z_2, z_3, z_4 and z_5 on a single Argand diagram.

b) Describe the mapping that transforms z_1 to z_2

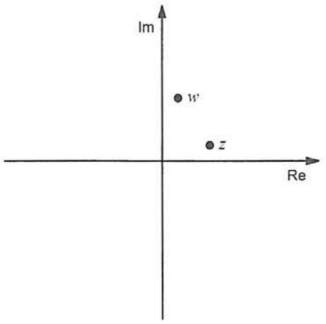
Question 14. Given that $z_1 = -5 + 2i$, $z_2 = -iz_1$, $z_3 = -iz_2$, $z_4 = -iz_3$ and $z_5 = -iz_4$

a) plot z_1, z_2, z_3, z_4 and z_5 on a single Argand diagram.

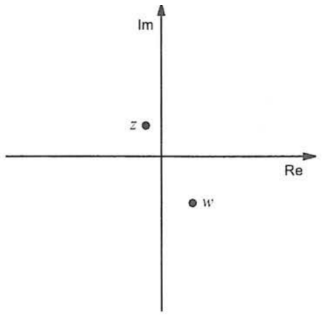
b) Describe the mapping that transforms z_1 to z_2

Question 15. For each of the following diagrams, copy the diagram and add the points z^*, z^2 and $z - w$

a)



b)



Question 16. In an Argand diagram the points representing the numbers $3 + 3i$ and $3 - i$ are two adjacent vertices of a square.

a) Find the area of the square.

b) Find all the possible pairs of numbers represented by the two other vertices.

Question 17. Distances on an Argand Diagrams

The distance between the origin and the complex number $a + bi$ is r where a and b are real constants.

a) Write r in terms of a and b

b) Show that $r^2 = |zz^*|$

Question 18. Two complex numbers are given by $z = a + bi$ and $w = c + di$ where a, b, c and d are real constants. The points z and w are a distance of p and q from the origin respectively.

a) Find zw

b) By considering the lengths in terms of a, b, c and d, prove that $|zw| = pq$

Question 19. Given that $z = 2 - 3i$

a) find iz

b) Find $(1 + i)z$

c) Prove that the shape formed by the origin, z, iz and $(1 + i)z$ is a square.

d) State the area of the square.

Question 20. It is given that $z = a + bi$ for constants a and b

a) Find iz

b) Find $(1 + i)z$

c) Prove that the shape formed by the origin, z, iz and $(1 + i)z$ is a square.

d) State the area of the square.

Question 21. The points on an Argand diagram P and Q are given by $z = a + bi$ and iz respectively, where a, b, c and d are real constants.

a) Find the length of OP and OQ

b) Prove that OP is perpendicular to OQ

Question 22. The points on an Argand diagram A, B and C are given by

$$z = 2 + i, \ z^3 \text{ and } 2z + z^3$$

respectively.

a) Plot the points A, B and C on an Argand diagram.

b) Find the length of the line segment OA

c) Find the length of the line segment BC

d) Describe the relationship between the length of the line segment OA and that of the line segment BC

Question 23. The points on an Argand diagram A, B and C are given by

$$z = 1 + 2i,\ z^* \text{ and } 3z + z^*$$

respectively.

a) Plot the points A, B and C on an Argand diagram.
b) Find the length of the line segment AC
c) Find the length of the line segment BC
d) Find the area of triangle ABC

Question 24. Given that $z = 1 + i$

a) find z^2, z^3, z^4 and z^5
b) Plot z, z^2, z^3, z^4 and z^5 on the same Argand diagram.
c) Find the distance between each of z^2, z^3, z^4 and z^5 and the origin
d) Hence, state the distance between z^n and the origin where n is an integer.

Question 25. Prerequisite knowledge for Q25: Complex Conjugates
It is given that

$$z = -20 + 21i \text{ and } w = 5 + 2i$$

a) Show that $\frac{z}{w} = -2 + 5i$
b) Plot w and $\frac{z}{w}$ on an Argand Diagram.
c) Prove that the origin and the points w and $\frac{z}{w}$ form a right-angled triangle.
d) Find the area of the triangle identified in part c).

Question 26. The points on an Argand diagram A and B are given by

$$z = 2 + 5i \text{ and } w = -2 + i$$

respectively.
Show that the area of triangle OAB is 6 units^2.

Question 27. The points on an Argand diagram A and B are given by $z = a + bi$ and $w = c + di$ respectively, where a, b, c and d are real constants.
Show that the area of triangle OAB is given by

$$\frac{1}{2} \times |\operatorname{Im}(zw^*)|$$

Complex Numbers: Roots of Equations

Questions 1–53

Question 1. A polynomial may have real roots or complex roots. For each of the following write down the possible combinations:

a) Cubic polynomial.
b) Quartic polynomial.
c) Quintic polynomial.

Question 2. Find all the roots of the equation

$$z^3 - 4z^2 + 30z - 52 = 0$$

given that one complex root is $1 + 5i$

Question 3. Find all the roots of the equation

$$-z^3 + 11z^2 - 64z + 170 = 0$$

given that one complex root is $3 - 5i$

Question 4. Given that

$$\mathrm{f}(z) = z^3 + az^2 + bz + c$$

where a, b and c are real numbers. It is given that $\mathrm{f}(z) = 0$ has solutions $z = 4$ and $z = 2 - 8i$
Find the values of a, b and c

Question 5. A cubic equation exists such that all its coefficients are real, state the equation given that two of its roots are -3 and $1+3i$

Question 6. Prerequisite knowledge for Q6: Argand Diagrams

A cubic f(z) has 3 distinct roots, which all lie on the same straight line in the complex plane. Given one root is $1+4i$

a) State the other two roots.

b) Find f(z)

Question 7. Given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + cz + d$$

where a, b, c and d are real numbers and $\mathrm{f}(z) = 0$ has roots $1 + i$, -2 and 3
Find the values of a, b, c and d

Question 8. A quartic equation $\mathrm{f}(z) = 0$ exists such that all its coefficients are real, find $\mathrm{f}(z)$ given that two of its roots are $1 - 5i$ and $2 + i$

Question 9. Given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + cz + d$$

where a, b, c and d are real numbers and that $\mathrm{f}(z) = 0$ has solutions $2 + i$ and $4 - 3i$
Find the values of a, b, c and d

Question 10. Find all the roots of the equation

$$z^3 - 13z^2 + 56z - 180 = 0$$

given that $z = 9$ is a solution.

Question 11. Find all the roots of the equation

$$z^3 - 10z^2 + 34z - 40$$

given that $(z - 4)$ is a factor.

Question 12. Given that $(z + 3)$ is a factor of $\mathrm{f}(z)$ where

$$\mathrm{f}(z) = z^3 + z^2 + 11z + 51$$

solve the equation $\mathrm{f}(z) = 0$

Question 13. Prerequisite knowledge for Q13-Q17: Argand Diagrams
Given that $z = 1 - 3i$ is a root of the equation

$$z^3 + 3z^2 + 50 = 0$$

Find the area of the triangle made by the three roots of the equation on an Argand diagram.

Question 14. Given that $z = 3$ is one solution of

$$z^3 - 7z^2 + 41z - 87 = 0$$

find the area of the triangle made by the three roots of the equation on an Argand diagram.

Question 15. Given that $z = 5 - \sqrt{3}i$ is a root of the equation

$$z^3 - 12z^2 + 48z - 56 = 0$$

a) find the area of the triangle made by the three roots of the equation on an Argand diagram.
b) Show that the triangle formed by the roots on the Argand diagram is equilateral.

Question 16. The roots of a quartic equation f(z) when plotted on an Argand diagram form a square of area 16 units^2. Each corner is in a different quadrant and one of the roots is $1 - 2i$

a) Sketch an Argand Diagram of all the solutions of f(z) = 0
b) Hence, determine f(z)

Question 17. The equation $\mathrm{f}(z) = 0$, where $\mathrm{f}(z)$ is a quartic polynomial with real coefficients, has a root $z = 3 + 5i$. Given that all the roots lie on a straight line in the complex plane and at least one of the roots is real:

a) Find the roots of $\mathrm{f}(z)$

b) Hence, determine $\mathrm{f}(z)$

Question 18. The function f is defined as

$$\mathrm{f}(z) = z^3 - 7z^2 + 32z - 60$$

a) Find $(2 + 4i)^3$

b) Verify that $2 + 4i$ is a root of $\mathrm{f}(z) = 0$

c) Express $\mathrm{f}(z)$ as a product of a linear factor and a quadratic factor with real coefficients.

Question 19. The function f is defined by

$$\mathrm{f}(z) = z^4 - z^3 - 6z^2 + 14z - 12$$

a) Show that $z = 1 + i$ is a root of $\mathrm{f}(z) = 0$
b) Find the remaining roots.

Question 20. Given that $z = 2 - 7i$ is root of the equation

$$z^4 - 6z^3 + 71z^2 - 146z + 530 = 0$$

find all the roots of the equation.

Question 21. The function f is defined by

$$\mathrm{f}(z) = z^4 + 5z^3 + 6z^2 + 50z + 100$$

It is known that $z = 1 + 3i$ is a solution to $\mathrm{f}(z) = 0$

a) Find all the solutions to $\mathrm{f}(z) = 0$

b) Hence, given the curve $y = \mathrm{f}(z)$ only has one turning point, sketch $y = \mathrm{f}(z)$

Question 22. The function f is defined by

$$\mathrm{f}(z) = z^4 + 4z^3 + 3z^2 + 10z + 50$$

a) Show that $1 + 2i$ is a root of $\mathrm{f}(z) = 0$

b) Find all the roots of $\mathrm{f}(z) = 0$

c) Hence explain why the graph of $y = \mathrm{f}(x)$ does not intersect the x-axis.

Question 23. Find in exact form, showing all your steps, all the solutions of

a) $z^3 - 13z^2 + 56z - 78 = 0$

b) $z^3 - 6z^2 + 15z - 14 = 0$

c) $z^3 - 2z^2 - 2z + 12 = 0$

d) $3z^3 - 20z^2 + 54z - 28 = 0$

Question 24. Prerequisite knowledge for Q24-Q26: Argand Diagrams

A cubic polynomial f(z) with real coefficients has 3 distinct roots, which all lie on the same straight line in the complex plane. It is known that one root is $1 + 4i$

a) State the other two roots.

b) Find f(z)

Question 25. A quintic polynomial f(z) with real coefficients has 1 real root and 4 complex roots which all lie on the same straight line in the complex plane. It is known that two of the roots are $2+3i$ and $2-2i$

a) State all the roots.

b) Find f(z)

Question 26. Given that $2+i$ and $1-3i$ are roots of

$$z^4 + az^3 + bz^2 + cz + d = 0$$

where a, b, c and d are real constants.

a) Plot all the roots on an Argand diagram.

b) Find the area formed on the Argand diagram by the roots.

c) Find the values of a, b, c and d

Question 27. A complex number α is given to be $1 + 2i$
A quartic polynomial f(z) with real coefficients exists such that it has roots α and α^2

a) Find α^2

b) Find f(z)

Question 28. Given that

$$z^3 - 4z^2 + kz - 10 = 0$$

where k is a real constant, has a root $1 - 2i$

a) Find all the roots.

b) Find the value of k

Question 29. Given that

$$z^3 + kz^2 - 5z - 12 = 0$$

where k is a real constant, has a root $z = 4$

a) Find the value of k

b) Find all the roots.

Question 30. It is known that $3 - i$ is a root of

$$2z^3 - 19z^2 + pz + q = 0$$

where p and q are real constants.

a) Find the values of p and q

b) Find all the roots.

Question 31. It is known that $2 - i$ is a root of

$$z^4 - 6z^3 + pz^2 + qz + 20 = 0$$

where p and q are real integers.

a) Express

$$z^4 - 6z^3 + pz^2 + qz + 20$$

as a product of two quadratic factors.

b) Hence find the values of p and q

Question 32. Given that $2i$ is a root of the equation

$$z^4 - 3z^3 - 6z^2 + pz + q = 0$$

where p and q are real constants.

a) Find the values of p and q

b) Find the other roots of the equation.

Question 33. It is given that one of the roots of

$$z^4 - 6z^3 + 2z^2 - 54z - 63 = 0$$

is a purely imaginary.

a) Without first finding any other roots, find the purely imaginary roots.

b) Hence, find all the roots to the equation.

Question 34. It is given that

$$\mathrm{f}(z) = z^4 + az^3 + bz^2 + az + 13$$

where a and b are real numbers, and $\mathrm{f}(z) = 0$ has a solution $3 - 2i$. Express $\mathrm{f}(z)$ as a product of two quadratic factors with real coefficients.

Question 35. Given that $-1 + \frac{\sqrt{3}}{2}i$ is a root of the equation

$$4z^4 + 16z^3 + 51z^2 + kz + 49 = 0$$

where k is an integer,

a) find all the roots.

b) determine the value of k

Question 36. A complex number α is given to be $2 - i$

a) Find α^2

b) Find α^3

It is known that for real values p and q,

$$p\alpha^3 + 5\alpha^2 + 12\alpha + q = 0$$

c) Find the values of p and q

d) Hence, solve the equation

$$pz^3 + 5z^2 + 12z + q = 0$$

Question 37. Prerequisite knowledge for Q37: Modulus-Argument form and Argand Diagrams
Given that

$$f(z) = z^2 - 3z + 9$$

a) Find the roots of $f(z) = 0$ giving your answer in modulus-argument form
b) Given that the roots of $f(z) = 0$ are also roots of $z^3 + k = 0$ where k is a real integer, find the other root of $z^3 + k = 0$
c) Plot all the roots of $z^3 + k = 0$ on an Argand diagram.

Question 38. It is given that

$$z^3 + kz^2 - kz - 1 = 0$$

where k is real.
a) Find a real root of the equation.
b) Find the range of values of k such that the equation has a pair of distinct complex roots.

Question 39. The polynomial f(z) has real coefficients. One solution of f(z) = 0 is $5 + 2i$

a) Show that

$$z^2 - 10z + 29$$

is a factor of f(z)

b) Find all the solution to f(z) = 0 given that

$$\mathrm{f}(z) = z^6 - 10z^5 + 29z^4 - 16z^2 + 160z - 464$$

Question 40. Prerequisite knowledge for Q40-Q46: Argand Diagrams

Given that $4 - 7i$ is a root of the equation

$$z^4 - 12z^3 + 117z^2 - 420z + 1300 = 0$$

Find the area of the quadrilateral made by the four roots of the equation on an Argand diagram.

Question 41. Find the area of the triangle when the three roots of the equation

$$z^3 - 12z^2 + 47z - 54 = 0$$

are plotted on an Argand Diagram.

Question 42. It is given that

$$z^4 + az^3 + bz^2 + cz + d = 0$$

where a, b, c and d are real numbers. When the roots of this equation are plotted on an Argand diagram they form a square with one vertex in each quadrant.
It is known that one root is $3 + 4i$

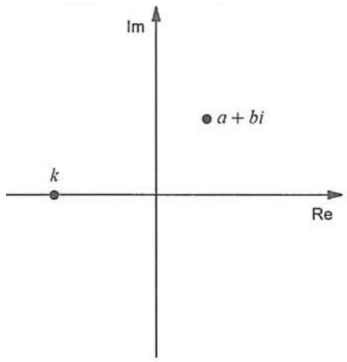
a) State the area of the square.
b) Determine the values of a, b, c and d

Question 43. For real numbers a and b the roots of

$$z^3 - 7z^2 + az + b = 0$$

form a triangle of area 56 when plotted on an Argand diagram.
Given that one of the roots is 3, find the values of a and b

Question 44. A cubic equation f(z) with real coefficients has two of its roots plotted on the Argand diagram below.
Find f(z) in terms of a, b and k



Question 45. Prerequisite knowledge for Q45: Complex Loci
Show that the roots of the equation

$$z^3 - 23z^2 + 299z - 2197 = 0$$

when plotted on an Argand diagram lie on the circle $|z| = k$, where k is a real number to be found.

Question 46. A cubic polynomial with real coefficients has 3 distinct roots. They are plotted on an Argand diagram and form an isosceles triangle with:

- an area of 42 units^2
- the distance between the complex roots which have real positive parts is 6 units
- the modulus of the complex roots is 5

Find the two possible cubic equations that satisfy these conditions.

Question 47. It is known that the equation

$$x^4 - x^3 + 8x^2 + 10x = 0$$

has a non-zero real root of a. Given that $a + 1 - 2i$ is a root of

$$z^4 + (a-2)z^2 + (13a - 15) = 0$$

Determine all the possible values of z

Question 48. The equation

$$z^4 - 8z^3 + 6z^2 + 148z - 272 = 0$$

has solutions $z = a$, $z = -b$ and $z = a + 3 + 3i$ where a and b are real positive integers.

a) State the fourth solution in terms of a

b) Find the value of a and the value of b

c) Hence state all the solutions to

$$x^4 - 8x^3 + 6x^2 + 148x - 272 = 0$$

Question 49. Prerequisite knowledge for Q49: Complex Loci
The function f is defined as

$$\mathrm{f}(z) = z^4 - 14z^3 + 74z^2 - 184z + 208$$

It is known that $\mathrm{f}(z) = 0$ has a root of $5 + i$

a) Solve $\mathrm{f}(z) = 0$

Two of the roots of $\mathrm{f}(z) = 0$ lie on the locus

$$C_1 = \{z : |z| = p\}$$

and the other two roots of $\mathrm{f}(z) = 0$ lie on the locus

$$C_2 = \{z : |z| = q\}$$

where $p > q$

b) Find the area of the region between C_1 and C_2 giving your answer in exact form.

Question 50. For each of the following, find all the roots and show that the sum of the roots of each of the following sum to zero:

a) $z^2 = 1$
b) $z^3 = 1$
c) $z^4 = 1$

Question 51. It is given that a cubic polynomial f(z) has real coefficients, and that the coefficient of the linear term is equal to the coefficient of the quadratic term. It has a complex root $p + qi$ and a real root k where p, q and k are real values.

a) Show that

$$k = -\frac{p^2 + q^2 + 2p}{1 + 2p}$$

b) Hence, given instead that f(z) has one real root and one repeated real root, show that

$$k = -\frac{p(p+2)}{2p+1}$$

Question 52. It is given that a cubic polynomial has real coefficients, and the linear term is zero.
Show that if it has a complex root $p + qi$ and a real root k where p, q and k are real values then $k^2 \geq q^2$

Question 53. It is given that a quartic polynomial has real coefficients, and the linear term is zero.
Show that if it has a complex root $p + qi$ and real roots m and n, where p, q, m and n are real non-zero constants, then

$$\frac{mn}{m+n} \geq q$$

Complex Numbers: Equations with Complex Coefficients

Questions 1–14

Question 1. Find an equation of the form

$$\mathrm{f}(z) = z^2 + pz + q$$

where p and q are complex numbers, given each of the following pairs of roots:

a) $z = 1 + i$ and $z = 3 + 2i$
b) $z = -3 + 4i$ and $z = 1 + 5i$
c) $z = -2 + 2i$ and $z = 2 - 2i$

Question 2. Find an equation of the form

$$\mathrm{f}(z) = z^3 + pz^2 + qz + r$$

where p, q and r are complex numbers, given that the solutions to $\mathrm{f}(z) = 0$ are $z = 1 + i$, $z = -1 + i$ and $z = 2 + 2i$

Question 3. A polynomial is defined by

$$f(z) = z^3 - (5+6i)z^2 + (35+27i)z - 58 - 30i$$

a) Find $(1+9i)^3$

b) Hence show that $z = 1+9i$ is a solution to $f(z) = 0$

It is known that a second solution to $f(z) = 0$ is $z = 2$

c) Find the remaining solution, writing your answer in the form $a+bi$ where a and b are integers to be found.

Question 4. Given that

$$f(z) = z^2 - (2+3i)z - k$$

where k is a complex number and $1+2i$ is a root of the equation $f(z) = 0$

a) Find the value of k

b) Show that $1-2i$ is not a root of the equation $f(z) = 0$

Question 5. It is known that $2 - i$ is a root of

$$\mathrm{f}(z) = z^3 - 3iz^2 + 2z + k$$

where k is a complex number

a) Find the value of k

b) Show that $2 + i$ is not a solution to $\mathrm{f}(z) = 0$

Question 6.

a) Find $\mathrm{f}(z)$ such that

$$z^3 - i = (z + i)\mathrm{f}(z)$$

b) Hence find all the solutions to $z^3 - i = 0$

Question 7. Find the solutions to

$$z^2 - (8 - 5i)z + 3 - 29i = 0$$

Question 8. The following questions should be solved by using methods relating the coefficients of a polynomial to sums and products of its roots

Given

$$z^2 - (10 - 2i)z + k = 0$$

where k is a complex number has roots a and $6 - 2i$. Find the value of k

Question 9. Given

$$z^2 - kz = -6 - 21i$$

where k is a complex number has roots a and $2 + 7i$. Find the value of k

Question 10. It is given that $1 + i$ and 3 are solutions of

$$z^3 - 2iz^2 + pz + q = 0$$

where p and q are complex numbers. Determine the values of p and q

Question 11. It is given that $2+i$ and -4 are solutions of

$$z^3 + pz^2 - (5+8i)z - q = 0$$

where p is a complex number and q is a real positive number. Find the values of p and q

Question 12. It is given that $2-3i$ and 3 are solutions of

$$2z^3 + pz^2 + qz - 13 = 0$$

where p and q are complex numbers. Determine the values of p and q

Question 13. The equation

$$z^3 - (10 - 2i)z^2 + pz + q = 0$$

where p and q are complex numbers has solutions $z = a$, $z = b$ and $z = a + bi$ for real values a and b

a) Find the values of a and b

b) Find the values of p and q

Question 14. Prerequisite knowledge for Q14: Argand Diagrams

The solutions of

$$z^3 - (6 + 3i)z^2 + (9 + 14i) - 15i = 0$$

all lie on the same straight line in the complex plane. It is known that two of the solutions are $z = 3$ and $z = 1 + 2i$ Find the remaining solution, writing your answer in the form $a + bi$ where a and b are integers to be found.

Complex Numbers: Modulus-Argument Form

Questions 1–54

Question 1. Given that $z = 4 - 2i$ and $w = -1 + 3i$, find

a) $|z|$
b) $|w|$
c) $|2w|$
d) $\left|\frac{z}{w}\right|$

Question 2. Find the exact value of each of the following:

a) $\arg(1 + i)$
b) $\arg(1 + \sqrt{3}i)$
c) $\arg(-2i)$
d) $\arg(-2 - 2i)$
e) $\arg(-3)$

Question 3. Write each of the following in the form $a + bi$ where a and b are real numbers:

a) $3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$
b) $\frac{1}{4}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$
c) $2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$
d) $7\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$
e) $12\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$
f) $\frac{\sqrt{2}}{4}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

Question 4. Write each of the following complex numbers into the form $a + bi$ where a and b are real numbers, giving your answers in exact form:

a) $|z| = 2$ and $\arg z = \frac{\pi}{3}$
b) $|z| = 1$ and $\arg z = \frac{\pi}{4}$
c) $|z| = 4$ and $\arg z = -\frac{2\pi}{3}$

Question 5. Write each of the following complex numbers in modulus-argument form. Give the argument in radians in exact form.

a) $3 + 3i$
b) $2i$
c) -4
d) $-1 + \sqrt{3}i$
e) $2 - 2i$
f) $-\sqrt{3} - i$
g) $-7 + 7i$
h) $3 - \sqrt{3}i$

Question 6. Write each of the following complex numbers in modulus-argument form, giving the argument to 3 significant figures.

a) $1 + 5i$
b) $-2 + 5i$
c) $4 - i$
d) $-3 - 7i$
e) $\sqrt{2} + 6i$
f) $-1 + \frac{3}{4}i$
g) $2 - \sqrt{3}i$
h) $-1 - \frac{1}{2}i$

Question 7. Ava is finding the principal argument form of $-\sqrt{3}-i$. She writes the following:

$$r=\sqrt{(-\sqrt{3})^2+(-1)^2}=\sqrt{3+1}=2$$

$$\tan\theta=\frac{-1}{-\sqrt{3}}=\frac{1}{\sqrt{3}}$$

$$\theta=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}$$

So $-\sqrt{3}-i=2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$

a) Identify and explain the mistake in Ava's working.

b) Find the correct modulus-argument form of $-\sqrt{3}-i$

Question 8. Write each of the following into principal argument form:

a) $4\left(\cos\frac{5\pi}{2}+i\sin\frac{5\pi}{2}\right)$

b) $2\left(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right)$

c) $3\left(\cos\frac{10\pi}{3}-i\sin\frac{10\pi}{3}\right)$

d) $-5\left(\cos\frac{10\pi}{3}+i\sin\frac{10\pi}{3}\right)$

Prerequisite knowledge for Q9-Q11: Complex Conjugates

Question 9. Express each of the following complex numbers in modulus-argument form, giving the argument correct to 3 significant figures:

a) $\frac{1}{1-3i}$
b) $\frac{1}{1+3i}$
c) $\frac{5}{2-3i}$
d) $\frac{-7+i}{7-2i}$

Question 10. State the complex conjugate of

$$z = 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

Question 11. It is given that $z = 4 - 5i$

a) Write z in modulus-argument form, giving the argument to 3 significant figures.

b) Hence state z^* in modulus-argument form.

Question 12. Determine the solutions to each of the following equations, giving your answer in modulus-argument form:

a) $z^2 - 2z + 2 = 0$

b) $9z^2 - 3z + 1 = 0$

c) $z^2 + 8z + 64 = 0$

d) $k^2z^2 - 2kz + 2 = 0$
where k is a positive real number.

Question 13. Given that $z_1 = 2 + 3i$ and $z_2 = iz_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 14. Given that $z_1 = 1 + 2i$ and $z_2 = 2iz_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 15. Given that $z_1 = 3 + 2i$ and $z_2 = (1 + i)z_1$

a) write z_1 and z_2 in modulus-argument form giving your answer to 3 significant figures.

b) Plot z_1 and z_2 on the same diagram.

c) Describe the geometrical transformation that maps z_1 onto z_2

Question 16.

a) Given $z_1 = 3 + 2i$ and $z_2 = 1 - 5i$, verify that $|z_1 + z_2| \leq |z_1| + |z_2|$

b) Explain geometrically why for two complex numbers z and w it is always true that

$$|z + w| \leq |z| + |w|$$

Question 17. It is known that

$$\arg(a+bi) = \frac{\pi}{4}$$

for real numbers a and b, find

a) $\arg(a-bi)$
b) $\arg(-a-bi)$
c) $\arg(b+ai)$
d) $\arg(b-ai)$

Question 18. Given that

$$\arg(a+bi) = -\frac{\pi}{3}$$

for real numbers a and b, find

a) $\arg(a-bi)$
b) $\arg(-a-bi)$
c) $\arg(b+ai)$
d) $\arg(b-ai)$

Question 19. Two complex numbers z and w on an Argand diagram are labelled A and B respectively. In each case, find the angle AOB giving your answer to 3 significant figures.

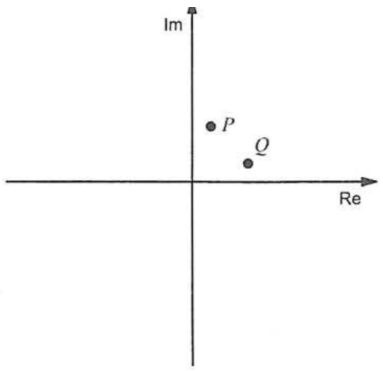
a) $z = 4 + 3i$, $w = 1 + 2i$
b) $z = 2 + 2i$, $w = -3 + 6i$
c) $z = -4 + 3i$, $w = 1 - i$

Question 20. Find the area of the triangle formed by vertices 0, z_1 and z_2 in each of these cases:

a) $z_1 = 2 + i$ and $z_2 = 1 + 3i$
b) $z_1 = 1 + 2i$ and $z_2 = -2 + 4i$
c) $z_1 = -5 + 3i$ and $z_2 = 2 - 3i$

Question 21. Given that $z_1 = 4 - i$, $z_2 = 6 - 8i$ and $z_3 = -4i$, find the exact area of the quadrilateral formed by the vertices of 0, z_1, z_2 and z_3

Question 22. Two complex numbers z_1 and z_2 are labelled as P and Q respectively.



a) On a copy of the Argand diagram, add a third point $z_1 + z_2$ and label the point R
b) Given that $OPRQ$ forms a rhombus, prove that $\arg(z_1 + z_2) = \frac{1}{2}(\arg z_1 + \arg z_2)$

Multiplying & Dividing in Modulus-Argument Form

Question 23. It is known that

$$z = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$w = 5\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) z^2
e) w^3

Question 24. It is known that

$$z = 2(\cos 1.4 + i\sin 1.4)$$

and

$$w = 4(\cos 0.7 + i\sin 0.7)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) w^2
e) w^2z
f) $\frac{z^2}{w^2}$

Question 25. It is known that

$$z = 7\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

and

$$w = 10\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $\frac{w}{z}$
d) $2w^2$

Question 26. For $z = 1 + i$ and $w = \sqrt{3} + i$, find zw^2 in principal argument form.

Question 27. Given that

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$w = 5\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right)$$

write the following complex numbers in principal modulus-argument form:

a) zw
b) $\frac{z}{w}$
c) w^3
d) $(1+i)z$

Question 28. The complex numbers z_1 and z_2 are given by

$$z = 2\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$$

and

$$w = \frac{1}{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) zw
b) $\frac{z}{w}$
c) $(zw)^2$
d) $(1+\sqrt{3}i)w$

Question 29. It is known that

$$z = \sqrt{3} + \sqrt{3}i \text{ and } w = 4i$$

Write the following complex numbers in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) z
b) w
c) Hence, find zw^2 in principal modulus-argument form.

Prerequisite knowledge for Q30: Complex Conjugates

Question 30. For the complex number z, it is known that $|z| = 3$ and $\arg z = \frac{2\pi}{3}$, find each of the following in **both** modulus-argument form and in the form $a + bi$ where a and b are real numbers:

a) z
b) z^*
c) z^2
d) $\frac{z}{z^*}$

Question 31. For the complex number z, in each case state the argument of z^2

a) $\arg(z) = \frac{\pi}{4}$

b) $\arg(z) = -\frac{\pi}{3}$

Question 32. In each case, express $\frac{z}{w}$ in the form $a + bi$ where a and b are real numbers given to 3 significant figures:

a) $|z| = 2$, $\arg z = \frac{\pi}{3}$ and $|w| = 3$, $\arg w = \frac{\pi}{4}$

b) $z = 1 + i$, $|w| = 2$ and $\arg w = \frac{\pi}{6}$

Question 33. The complex number z is given by $5 - 2i$ and another complex number w exists such that

$$wz = \sqrt{290}(\cos 0.8 + i \sin 0.8)$$

a) Find $|w|$
b) Find $\arg w$ giving your answer to 3 significant figures.

Question 34. Given that

$$z_1 = -\frac{\sqrt{3}}{2} - \frac{3}{2}i$$

$$|z_1 z_2| = \sqrt{30}$$

and

$$\arg(z_1 z_2) = \frac{\pi}{4}$$

a) Write z_1 in the form $r(\cos\theta + i\sin\theta)$
b) Find z_2 in the form $a + bi$ where a and b are real numbers.

Question 35. Two complex numbers, z and w may be represented on an Argand diagram where

$$\text{Im}(z) > \text{Im}(w)$$

The angle between them is $\frac{\pi}{3}$ radians and it is known that $|3z| = |w|$
Find $\frac{w}{z}$ in the form $a + bi$ where a and b are exact real numbers.

Question 36. The complex number z is

$$-k + ki$$

where k is a positive integer.

a) Express z is modulus-argument form.

b) Given that

$$w = 2\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

and

$$\text{Re}(zw) = -20$$

find the value of k

Question 37. Given that

$$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

find iz^3 writing your answer in principal modulus-argument form.

Prerequisite knowledge for Q38: Compound Angle formulae

Question 38. Prove that

a)

$$|z_1 z_2| = |z_1||z_2|$$

and

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

b)

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

Question 39. Prove that, given z is non-zero,

$$\arg\left(\frac{1}{z}\right) = -\arg z$$

Question 40. Given that

$$z = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

find iz^2 writing your answer in principal modulus-argument form.

Question 41. It is given that $z = a + bi$ where a and b are real constants. It is additionally known that

$$|z^6| = 729 \text{ and } \arg(z^6) = \frac{5\pi}{3}$$

Determine the value of a and b giving your answer in exact form.

Question 42. On the Argand diagram, the points A and B are given by

$$z = 2\sqrt{3} + 2i \text{ and } w = -\sqrt{2} + \sqrt{2}i$$

respectively.
By writing z and w in modulus-argument form, find the geometrical transformations that map A onto B

Question 43. On the Argand diagram, the points A and B are given by

$$z = 1 + 4i \text{ and } w = -3 + 2i$$

respectively.
Find the multiplication scale factor and angle of rotation that maps A onto B

Question 44. On the Argand diagram, the points A and B are given by

$$z = 4 + 2i \text{ and } w = 1 - 5i$$

respectively. Find the multiplication scale factor and angle of rotation that maps A onto B

Question 45. For two complex numbers, z and w it is known that

$$\arg(z) = \frac{\pi}{6} \text{ and } \arg(w) = \frac{2\pi}{3}$$

Sketch and label the points representing z, w and zw on an Argand diagram for each of the following cases:

a) $|z| = 2$, $|w| = 3$

b) $|z| = 4$, $|w| = 1$

Question 46. For two complex numbers, z and w it is known that

$$|z| = 2 \text{ and } |w| = 1$$

Sketch and label the points representing z, w and zw on an Argand diagram for each of the following cases:

a)

$$\arg(z) = \frac{\pi}{4} \text{ and } \arg(w) = \frac{2\pi}{5}$$

b)

$$\arg(z) = \frac{\pi}{6} \text{ and } \arg(w) = -\frac{5\pi}{8}$$

Question 47. Given that

$$z_1 = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

and

$$z_2 = 3\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

a) sketch z_1 and z_2 on the same Argand diagram, labelling the points A and B respectively.

A third complex number z_3 which satisfies $|z_3| = 3$ and $-\pi < \arg(z_3) < 0$ and is plotted on the same Argand diagram as point C

b) Find the angle ACB

Question 48. It is known that z and w are complex numbers.

a) Show that $|z|^2 = zz^*$

b) Show that $(z - w)^* = z^* - w^*$

c) Find $|z - w|^2$ in terms of z, z^*, w and w^*

It is given that $|z + w| = 4$, $|z| = 2$ and $|w| = 3$

d) Find $|z - w|$

Question 49. It is known that z and w are complex numbers. Show that

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2$$

Question 50. Given

$$z = k(\cos\theta + i\sin\theta)$$

where k is real,

a) show that $|z| = k$

b) Find $\arg(z+k)$ in terms of θ

Question 51. The complex numbers z and w are such that

$$z = 2 + 2\sqrt{3}i$$

and

$$w = 1 + i$$

a) Find $\frac{z}{w}$ in the form $a + bi$
b) Write z in the form $r(\cos\theta + i\sin\theta)$
c) Write w in the form $r(\cos\theta + i\sin\theta)$
d) Find $\frac{z}{w}$ in the form $r(\cos\theta + i\sin\theta)$
e) Hence show that

$$\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

Question 52. The complex numbers z and w are such that

$$z = 4\sqrt{3} - 4i$$

$$\arg(w) = \frac{\pi}{4} \text{ and } |w| = 4$$

a) Find w in the form $a + bi$
b) Hence find $\frac{z}{w}$ in the form $a + bi$
c) Write z in the form $r(\cos\theta + i\sin\theta)$
d) Find $\frac{z}{w}$ in the form $r(\cos\theta + i\sin\theta)$
e) Hence show that

$$\cos\left(-\frac{5\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Question 53. For each of the following cases, find the smallest value of n such that $\arg((wz)^n) = 0$ and state $|(wz)^n|$ for that value of n:

a) $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, $w = 5\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

b) $z = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, $w = 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

c) $z = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$,

$$w = 3\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

Question 54. Given that n is a positive integer and

$$\left(\frac{1+\sqrt{3}i}{2-2i}\right)^n$$

is a real number, find the smallest possible value of n

Complex Numbers - Loci in the Argand Diagram

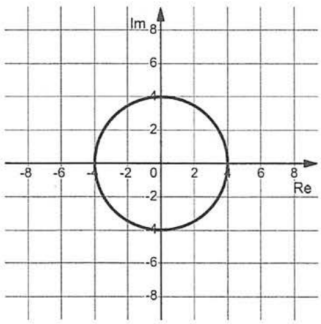
Questions 1–40

Question 1. Sketch the locus of z on a separate Argand diagram in each case, including identifying the maximum and minimum real and imaginary values of z:

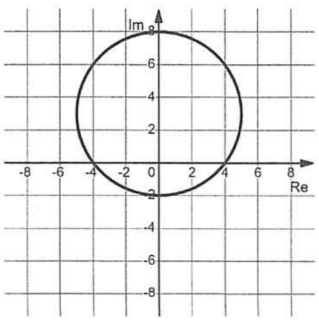
a) $|z| = 6$
b) $|z - 1| = 5$
c) $|z - 2i| = 4$
d) $|z - (4 + 2i)| = 2$
e) $|z - (1 - 3i)| = 5$
f) $|z - 1 + i| = 6$
g) $|z - 4 + 2i| = \sqrt{10}$
h) $|z + 3 - 4i| = \sqrt{5}$

Question 2. For each of the following Argand diagrams, identify the locus of z:

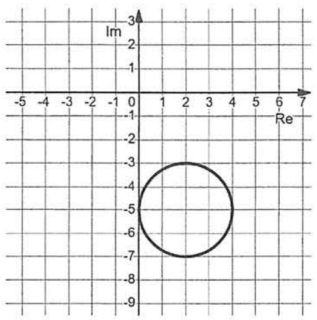
a)



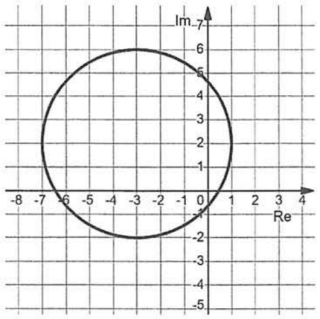
b)



c)



d)



Question 3. For each of the following, find the Cartesian equation of the locus of z and sketch the locus of z, identifying any values where the locus intersects with the axes:

a) $|z+3|=2$
b) $|z+5i|=6$
c) $|z-(-3-i)|=1$
d) $|z-(-1+4i)|=\sqrt{3}$
e) $|z+2-6i|=7$
f) $|1+2i-z|=4$

Question 4. It is known that the complex number z satisfies

$$|z-1-4i|=7$$

a) Sketch the locus of z on an Argand diagram.
b) Find the exact values for which both $|z-1-4i|=7$ and $\mathrm{Re}(z)=0$
c) Find the exact values for which both $|z-1-4i|=7$ and $\mathrm{Im}(z)=0$

Question 5. It is known that the complex number z satisfies

$$|z - 4 - 4i| = 3$$

a) Sketch the locus of z on an Argand diagram.
b) Find the Cartesian equation of this locus.
c) Find the maximum value and minimum value of $\arg(z)$ in the interval $(-\pi, \pi]$ giving your answer to 3 significant figures.
d) Find the maximum and minimum values of $|z|$ for points on the locus.

Question 6. It is known that the complex number z satisfies

$$|z - 3 + 5i| = 2$$

a) Sketch the locus of z on an Argand diagram.
b) Find the Cartesian equation of this locus.
c) Find the maximum value and minimum value of $\arg(z)$ in the interval $(-\pi, \pi]$ giving your answer to 3 significant figures.
d) Find the maximum and minimum values of $|z|$ for points on the locus.

Question 7.

a) Sketch the two loci given by

$$|z-5|=3 \text{ and } |z-1|=5$$

on the same Argand diagram.

b) Find the point of intersection of the two loci in the form $a+bi$ where a and b are real values.

Question 8. Two complex numbers are given by

$$z_1=-2+4i \text{ and } z_2=6-2i$$

It is known that z_1 and z_2 are the ends of a diameter of a circle.

a) Find the equation of the circle in the form

$$|z-a-bi|=r$$

where a, b and r are real values to be found.

There is a complex number w which satisfies

$$|w-p-qi|=4$$

It is known that $6+4i$ is the only complex number which satisfies both loci and that for all points on the locus, $\text{Re}(w)>0$ and $\text{Im}(w)>0$

b) Find the values of p and q

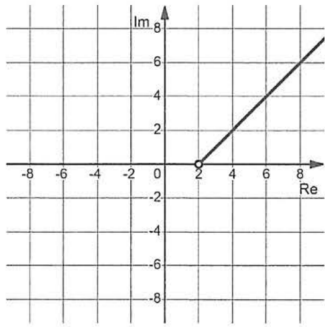
Half-lines

Question 9. Sketch the locus of z on a separate Argand diagram in each case, without identifying any values at which the locus intersects with the real or imaginary axes:

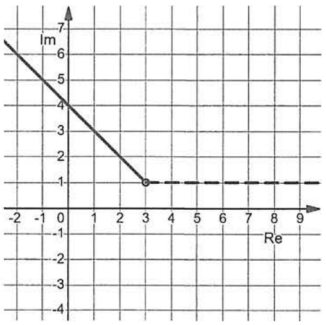
a) $\arg(z) = \frac{\pi}{6}$
b) $\arg(z - 3i) = \frac{\pi}{9}$
c) $\arg(z - 4) = -\frac{\pi}{4}$
d) $\arg(z - (2 + i)) = \frac{3\pi}{4}$
e) $\arg(z - (2 - 4i)) = -\frac{\pi}{3}$
f) $\arg(z - 5 + 3i) = \frac{\pi}{5}$
g) $\arg(z + 1 + 2i) = -\frac{\pi}{5}$
h) $\arg(z + 3 - 2i) = -\frac{3\pi}{5}$

Question 10. For each of the following Argand diagrams, identify the locus of z:

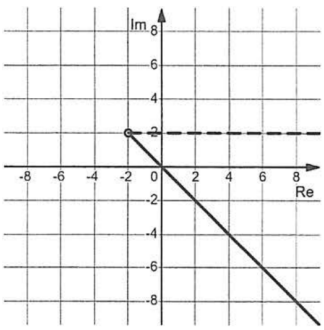
a)



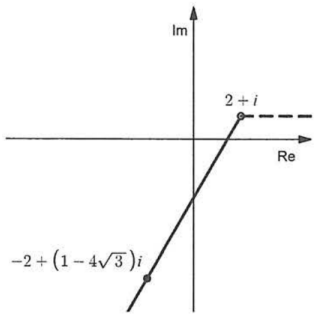
b)



c)



d)



Question 11. Sketch the locus of z, identifying any values where the locus intersects with the real or imaginary axes and find the Cartesian equation of the locus of z for:

a) $\arg(z) = \frac{\pi}{2}$
b) $\arg(z+2) = -\frac{\pi}{4}$
c) $\arg(z+i) = \frac{2\pi}{3}$
d) $\arg(z-(-2-5i)) = -\frac{\pi}{6}$
e) $\arg(z-(-3+3i)) = \frac{7\pi}{8}$
f) $\arg(z+2-4i) = -\frac{\pi}{9}$

Question 12. Given that the complex number z satisfies both

$$\arg(z) = \frac{\pi}{4} \text{ and } \arg(z-3) = \frac{\pi}{2}$$

a) Sketch the locus of $\arg(z) = \frac{\pi}{4}$ and $\arg(z-3) = \frac{\pi}{2}$ on the same diagram.
b) Find z in the form $a+bi$
c) Hence, find $\arg(z-2i)$ giving your answer to 3 significant figures.

Question 13. It is known that the complex number z satisfies

$$\arg(z + 2i) = \frac{\pi}{3}$$

a) Sketch the locus of z on an Argand diagram.

The complex number w lies on the locus of z given above such that $|z|$ is a minimum.

b) Find w in the form $a + bi$ where a and b are real values.

c) Give the minimum value of $|z|$

Question 14. The complex number z satisfies $\arg(z + 6i) = \frac{\pi}{4}$ Additionally, it is given that $w = 1 + 2i$

a) Sketch the locus of z and the point w on an Argand Diagram.

b) Find the minimum distance between w and the locus of z giving your answer in exact form.

Question 15. The complex number z satisfies $\arg(z-2+2i)=\frac{2\pi}{3}$. Given also $w=2\sqrt{3}+2i$

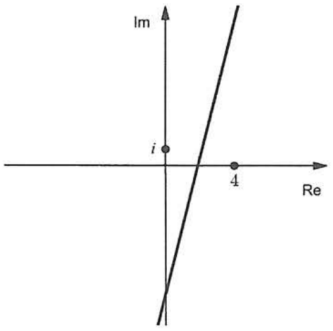
a) sketch the locus of z and the point w on an Argand Diagram.

b) Find the minimum distance between w and the locus of z giving your answer in the form $a+b\sqrt{3}$ where a and b are integers to be found.

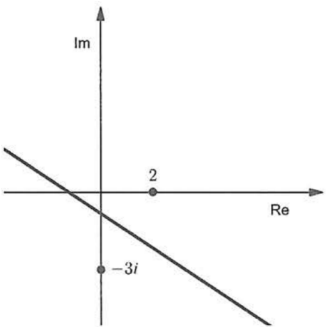
Lines

Question 16. For each of the following Argand diagrams, identify the locus of z using the identified points, and find the cartesian equation of the locus of z in the form $y=mx+c$:

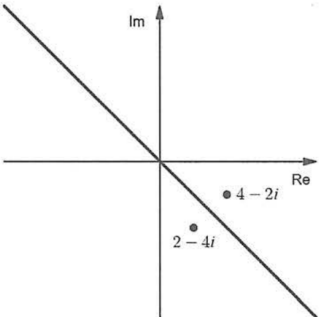
a)



b)



c)



Question 17. Sketch the locus of z, identifying any values where the locus intersects with the real or imaginary axes and find the Cartesian equation of the locus of z in the form $y = mx + c$, for each of the following:

a) $|z - 2| = |z - 4|$
b) $|z + 3 + 5i| = |z + 3 - 2i|$
c) $|z - 2| = |z - i|$
d) $|z - 2| = |z + 3i|$
e) $|z - (-5 - i)| = |z + (2 - 3i)|$
f) $|z + 5 - 3i| = |z - 4 + 7i|$

Question 18.

a) Sketch the two loci given by

$$|z - 2 + i| = |z - 3 + 2i|$$

and

$$|z - 1| = |z + 4i|$$

on the same Argand diagram.

b) Find the value of the complex number z which satisfies both of the loci above, in the form $a + bi$ where a and b are real values.

Question 19. It is known that the complex number z satisfies

$$|z+4| = |z-7i|$$

a) Sketch the locus of z on an Argand diagram. Label any intersections with the real and imaginary axes.
b) Find the Cartesian equation of this locus.
c) Find the least possible value of $|z|$ giving your answer in exact form.

Question 20. It is known that the complex number z satisfies

$$|z-2+3i| = |z-4+4i|$$

a) Sketch the locus of z on an Argand diagram. Label any intersections with the real and imaginary axes.

Furthermore, the complex number z satisfies

$$\arg(z) = -\frac{\pi}{4}$$

b) Find z in the form $a+bi$

Question 21.

a) Show on Argand diagram the solutions to

$$|z+1| = |z-6| = |z-2i|$$

b) Hence find the complex number z which satisfies $|z+1| = |z-6| = |z-2i|$

Mix

Question 22. In each case, sketch on an Argand diagram the locus of z given both conditions, label the possible value(s) of z:

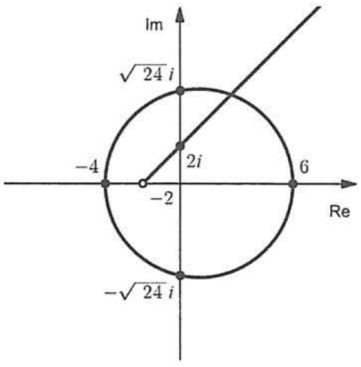
a) $|z| = 3$ and $\arg(z+2) = \frac{\pi}{4}$
b) $|z-3-3i| = 3$ and $\arg(z-3-3i) = \frac{\pi}{3}$
c) $|z-3i| = |z-5i|$ and $\arg(z-1-i) = \frac{3\pi}{4}$
d) $|z+5+2i| = 4$ and $\arg(z-1) = -\frac{3\pi}{4}$
e) $|z+1| = |z+5|$ and $\arg(z+2i) = -\frac{2\pi}{3}$
f) $\arg(z-2+i) = \frac{\pi}{6}$ and $|z-3i| = 2$
g) $|z-2+2i| = 1$ and $|z-2| = |z-2+4i|$
h) $|z+1| = 4$ and $|z+1-i| = \left|z+2+\frac{1}{2}i\right|$
i) $\arg(z-2) = \frac{\pi}{4}$, $|z-1| = |z-i|$ and $|z-2-2i| = 5$

Question 23. Show on Argand diagram the solutions to

$$|z-3| = |z+1| = 4$$

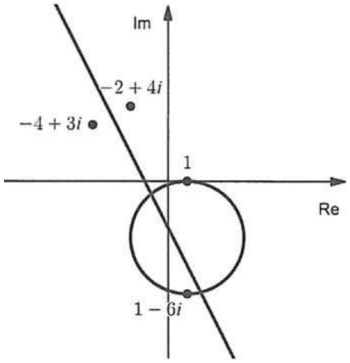
Label the two solutions P and Q

Question 24. For the diagram below:



a) Find the complex locus such that a complex number z_1 lies on the circle in the diagram above.
b) Find the complex locus such that a complex number z_2 lies on the half-line in the diagram above.
c) Find the complex number z such that z satisfies both complex loci above, giving your answers in the form $a+bi$

Question 25. For the diagram below:



a) Find the complex locus such that a complex number z_1 lies on the circle in the diagram above.
b) Find the complex locus such that a complex number z_2 lies on the line in the diagram above.
c) Find the complex numbers which satisfy both complex loci above, giving your answers in the from $a + bi$

Question 26. By first sketching the two loci on the same Argand diagram, find the value of the complex number z such that

$$\left\{z : |z - i| = 2\sqrt{3}\right\} \cap \left\{z : \arg(z + 2 - i) = \frac{\pi}{3}\right\}$$

Question 27. By first sketching the two loci on the same Argand diagram, find the value of the complex number z such that

$$\{z : |z| = 4\} \cap \left\{z : \arg(z - 4\sqrt{2}) = \frac{3\pi}{4}\right\}$$

Question 28. It is known that the complex number z satisfies

$$|z - 3 - 3i| = 3$$

a) Sketch the locus of z on an Argand diagram.

b) Find $|z|$ given that z satisfies

$$|z - 3 - 3i| = 3$$

and

$$\arg(z - 6) = \frac{3\pi}{4}$$

giving your answer in exact form.

Question 29. It is known that the complex number z satisfies

$$|z-2|=4$$

a) Sketch the locus of z on an Argand diagram, label any points that intersect with the real and imaginary axes.

It is also known that z satisfies

$$\arg(z-2)=-\frac{\pi}{4}$$

b) Express z in the form $a+bi$ where a and b are real numbers.

c) Find $|z|^2$ in its simplest form.

Question 30. Sketch the locus of the complex number z given

$$|z-a-bi|=k$$

where $a=b$, $a>0$ and $k<a$

In each case, label the point(s) z given additionally that:

a) $\arg(z-a-bi)=\frac{\pi}{3}$

b) $\arg(z-a-bi)=-\frac{\pi}{4}$

c) $|z-a|=|z-bi|$

d) $|z|=a$

e) $\arg(z)$ is a maximum

f) $\arg(z)$ is a minimum

Question 31. It is known that the complex number z lies on the locus given by

$$\arg(z - 2i) = \frac{\pi}{4}$$

a) Sketch the locus of z on an Argand diagram.
b) Find an equation of a circle with centre $4 + 3i$ that touches the locus in part a), giving your answer in complex form.

Question 32. The complex number z satisfies

$$|z - 2 - 3i| = |z - 5|$$

and

$$|z - ki| = 2$$

where k is a real number. Find the interval in which k must lie.

Question 33. The complex number z satisfies

$$\arg(z-3-ki)=\frac{3\pi}{4}$$

and

$$|z-1-ki|=|z-2|$$

where k is a positive real number. Find the interval in which k must lie.

Question 34. It is known that the complex number z satisfies

$$|z-2-3i|=k$$

and

$$\arg(z-4)=\frac{3\pi}{4}$$

where k is a real number. Find the exact range of values of k

Question 35. Given that the complex number z satisfies

$$|z - 6 - 5i| = 1$$

a) Sketch the locus of z on an Argand diagram.
b) Show that the maximum value of

$$\theta = \arg(z - 2 - 4i)$$

for $-\pi < \theta \leq \pi$ is $\arctan\left(\frac{8}{15}\right)$

Question 36. Given that the complex number z satisfies

$$|z + 4 - 6i| = 3$$

a) Sketch the locus of z on an Argand diagram.
b) Show that the minimum value of

$$\theta = \arg(z - 3 - 3i)$$

for $-\pi < \theta \leq \pi$ is $\frac{\pi}{2} + \arctan\left(\frac{20}{21}\right)$

Question 37. Given that the complex number z satisfies

$$|z + 3 - 5i| = 2$$

a) Sketch the locus of z on an Argand diagram.
b) Find the range of values of θ for which z also satisfies $\theta = \arg(z - 2 - 3i)$ where $\pi < \theta \leq \pi$ giving your answer to 3 significant figures.

Question 38. Given that the complex number z satisfies

$$|z + 1| = 4 \text{ and } \theta = \arg(z - 1)$$

where $-\pi < \theta \leq \pi$
a) find the values of θ such that z additionally satisfies $|z - 2| = 2$
b) Hence state the range of values of θ such that $|z - 2| < 2$

Question 39. The complex number z satisfies

$$|z| = |z - k^2 + 4i|$$

and

$$\arg(z - 2k + 2i) = \frac{\pi}{3}$$

where k is a real number.

a) Show that

$$\mathrm{Re}(z) = \frac{k^4 - 16\sqrt{3}k}{2k^2 - 8\sqrt{3}}$$

b) Find $\mathrm{Re}(z)$ when $\mathrm{Im}(z) = 0$ giving your answer to 3 significant figures. State the value(s) of k to 3 significant figures for when these complex numbers occur.

c) Explain with reference to the loci why the solution to your part a) is not valid when $k = 0$ and $k = 4$

d) Find the range of values of k such that z exists.

Question 40.

a) For $z = x + iy$, show that

$$\frac{z-4}{z} = \frac{x^2 - 4x + y^2}{x^2 + y^2} + \frac{4y}{x^2 + y^2}i$$

It is known that the complex number z satisfies

$$\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$$

b) Show that the equation $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$ describes the locus of part of a circle.

c) Sketch the locus of z in the Argand diagram.

Complex Numbers: Loci - Inequalities

Questions 1–15

Question 1. In each case, shade the region on the Argand diagram for the complex number z, labelling any points where loci intersect with the axes:

a) $|z| \geq 2$
b) $|z + 2i| < 4$
c) $|z - 3 + i| \leq 1$
d) $|z + 1 - i| > 5$
e) $2 \leq |z| \leq 4$
f) $|z - 1 - i| \geq 1$ and $|z - 2 - 2i| \leq 2$

Question 2. In each case, shade the region on the Argand diagram for the complex number z, labelling any points where loci intersect with the axes:

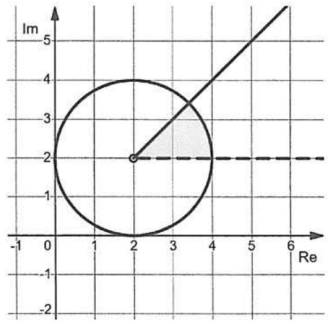
a) $0 < \arg(z - 2i) < \frac{\pi}{4}$
b) $\frac{\pi}{3} \leq \arg(z + 1) \leq \frac{2\pi}{3}$
c) $-\frac{\pi}{4} < \arg(z - 1 + 3i) \leq \frac{\pi}{3}$
d) $-\frac{\pi}{6} \leq \arg(z + 2 - i) < \frac{\pi}{6}$
e) $\arg(z - 4 + 2i) \leq \frac{2\pi}{3}$ and $\arg(z + 2 - i) \geq \frac{\pi}{4}$

Question 3. In each case, shade the region on the Argand diagram for the complex number z, labelling any points where loci intersect with the axes:

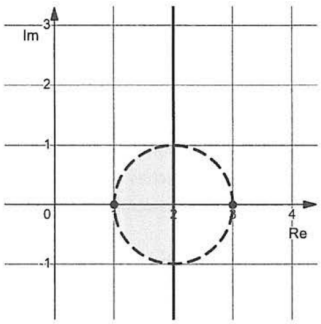
a) $|z-2| > |z-5|$
b) $|z-3| \geq |z+4i|$
c) $|z+1-2i| < |z-4i|$
d) $|z-4-3i| \leq |z+2-8i|$
e) $|z-1| > |z+3|$
and $|z-3-2i| \geq |z+1-4i|$

Question 4. For each of the following Argand diagrams state the conditions for the set of points z represented by the shaded region. Give your answer in set notation.

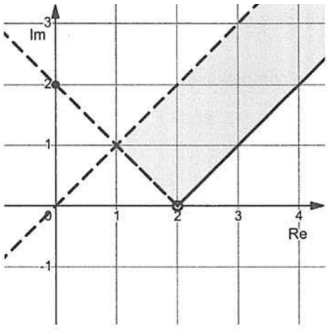
a)



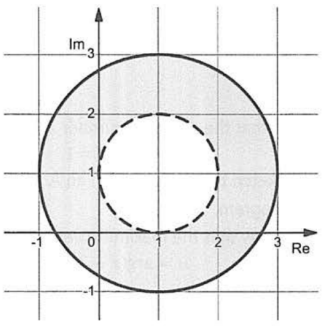
b)



c)



d)



Question 5.

a) On separate Argand diagrams shade the regions represented by $|z-3-4i| \leq 2$ and $0 \leq \arg(z-2-i) \leq \frac{\pi}{4}$

b) Hence shade the region satisfied by

$$\{z \in \mathbb{C} : |z-3-4i| \leq 2\}$$
$$\cap \left\{z \in \mathbb{C} : 0 \leq \arg(z-2-i) \leq \frac{\pi}{4}\right\}$$

Question 6.

a) On separate Argand diagrams shade the regions represented by $\left|z-2-\frac{1}{2}i\right| < 4$ and $\frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}$. You do not need to label intersections with the axes.

b) Determine whether $2+\frac{1}{2}i$ lies in the region $\frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}$

c) Hence, shade on an Argand diagram, the set of points

$$\left\{z \in \mathbb{C} : \left|z-2-\frac{1}{2}i\right| \leq 4\right\}$$
$$\cap \left\{z \in \mathbb{C} : \frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}\right\}$$

Question 7.

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 4 - 3i| \leq 2\}$$

b) Find the greatest and least value of $\arg z$ giving your answer in exact form.

Question 8.

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z + 4 + 4i| \leq 2\}$$

b) Find the greatest and least value of $\arg z$ giving your answer to 3 significant figures.

Question 9. For the complex number z find the maximum value of $|z|$ such that

a) $|z-4i| \leq 4$ and $0 \leq \arg z \leq \frac{\pi}{3}$

b) $|z-2| \leq 4$ and $\frac{\pi}{4} \leq \arg z \leq \frac{3\pi}{4}$

c) $|z-1-i| \leq 3$ and $|z-1-i| \geq |z+1-i|$

Question 10. For the complex number z find the minimum value of $|z|$ such that

a) $|z-2-4i| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{4}$

b) $|z-3-3i| \leq \sqrt{2}$
and $\frac{\pi}{4} \leq \arg(z-1-i) \leq \frac{3\pi}{4}$

c) $|z-2| \leq 3$ and $|z+7-5i| \leq |z+3+7i|$

Question 11.

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 2 - 2i| \leq 6\}$$

Label any intersects with the axes.

b) Determine whether $9\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ lies in the region.

c) Determine whether $4\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$ lies in the region.

Question 12.

a) Shade on an Argand diagram, the set of points

$$\left\{z \in \mathbb{C} : \tan^{-1}\left(\frac{3}{4}\right) \leq \arg(z + 3i) \leq \frac{\pi}{2}\right\}$$

$$\cap\{z \in \mathbb{C} : |z - 1 + 2i| \leq 1\}$$

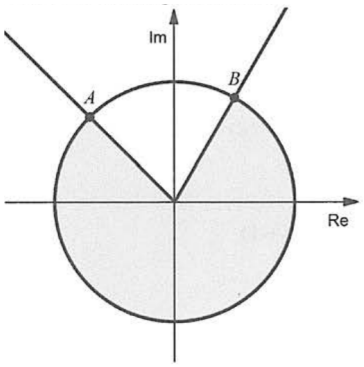
b) Find the value at which $|z|$ is a minimum.

Prerequisite knowledge for Q13-Q15: Sector Area in Radians

Question 13. A region may be defined by

$$\{z \in \mathbb{C} : |z| \leq 5\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{3} \leq \arg z \leq \frac{3\pi}{4}\right\}$$

as shown in the diagram below:



a) Find the complex numbers representing points A and B on the diagram, giving your answer in exact form.
b) Find the area of the shaded region.

Question 14.

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 4 - 3i| \leq 4\}$$

$$\cap\{z \in \mathbb{C} : |z - 2| \leq |z - 10|\}$$

b) Show that the area of the shaded region is

$$\frac{32}{3}\pi + 4\sqrt{3}$$

Question 15. Find the area of the region defined by

$$\{z \in \mathbb{C} : |z| \leq 2\} \cap \{z \in \mathbb{C} : |z - 3i| \leq 3\}$$

Give your answer to 3 significant figures.